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# A general QoE assessment framework for applications and services

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ABSTRACT

In this paper, we have designed a testbed to evaluate end-to-end Quality of Experience (QoE) performance for mobile applications and devices in a controllable and repeatable manner. Various network scenarios have been defined which allow replication of real-life conditions in a controlled environment. The general testbed framework is presented, which is comprised of a deconstruction process and an aggregation process. A convergence study of the testbed has been carried out to identify the required number of iterations for achieving stable results, and human panel experiments have been introduced to calibrate the performance of the testbed. A case study of two use cases, i.e. content streaming and high-speed internet, is presented with focus on the user experience domain. The measurement results have demonstrated the measurement capability of the proposed testing methodology, which is flexible to generalize the QoE computations for many of the use cases. It is shown that with the help of the testbed, we can better understand how radio resources and propagation channel models will influence the performance of the application/service under test.

## 1. Introduction

5G is envisioned to bring a variety of new services and applications (e.g., video streaming, virtual reality, social networking, gaming, etc.) to the end user with faster transmissions, massive connectivity, lower latency, and higher reliability. One crucial task is to ensure that the services provided by 5G will work and fulfill certain requirements experienced by the end user in certain network scenarios. Understanding the end users' expectations and experience is of paramount importance to service providers and application developers, as it helps to improve the service quality perception with the aim to place themselves at an advantageous position in a highly competitive service provision market.

Traditionally Quality of Service (QoS) is the most widely used service quality metric, which can be used to assess the performance of telecommunication services by performing a series of technical measurements such as throughput, delay, jitter, error probability, packet loss, etc. Although QoS has been used to indicate the effectiveness of a service, it does not take user's subjective factors into account, which directly reflect the user perceived degree of services. Quality of Experience (QoE), defined by the ITU-T as *"overall acceptability of an application or service, as perceived subjectively by the end-user"* [1], on the other hand is multidimensional in nature, including both objective, technology-oriented aspects, and subjective, user-oriented aspects. The former pays attention to the influence of user perception based on application and network performance, while the latter is focusing on individual user experience from user's cultural background, psychological profiles, social-economic issues, and so on.

QoE assessment is important to ensure that the provided services can meet the end user's expectation and satisfaction, and to identify the key factors that may influence the user perceived QoE. However, the QoE assessment in an End-to-End (E2E) communication system is a complex task, as it depends on many factors, such as the network conditions including both radio access network and core network, the terminal capabilities, the service providers, and human factors. The combination of various use cases and network configurations further complicates the situation and makes the theoretical evaluation of QoE for a given 5G service impossible.

Typically QoE can be evaluated via subjective methods where a group of end users rate the perceived quality using the mean opinion score (MOS) [2]. The disadvantage of subjective methods is that it is laborious, time consuming and typically can be conducted offline. Objective methods like field testing on the other hand, utilize the QoE model with a large number of measured parameters to predict user perceived QoE. But it is very costly and the results are unpredictable

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given the lack of control over the testing environment, e.g., changing radio propagation conditions, network conditions, etc. Therefore, it is of great interest to design a fully lab controlled E2E testbed with a homogeneous testing framework that allows E2E QoE evaluation for new mobile applications and services in an automated and repeatable manner.

This paper presents part of the work in the H2020 TRIANGLE project [3]. The aim of the project is to design an E2E fully lab-controlled testbed, which is built on top of the FIRE project by extending and enhancing the PerformNetworks [4] testbed to cover 5G scenarios. The testbed has full control of all elements between the service provider and the commercial handset with different brands and operating systems. In between the two ends, an emulated radio access network and a full commercial core network are configured and can be controlled automatically to enable extensive laboratory testing of services against different scenarios and configurations [5]. The TRIANGLE framework has been accepted by Next Generation Mobile Networks (NGMN) Alliance [6].

The approach to assess the QoE of mobile applications using the TRIANGLE testbed was presented in our earlier work [7]. Comparatively, this article provides a more comprehensive description of the testing methodology which is flexible to generalize the QoE computation for many of the applications and services. One of the most important tasks before executing any test campaign is the calibration of the testbed, which includes the calibration of the measurements as well as the Extract, Transform and Load (ETL) process. Additionally, two example use cases have been presented to demonstrate the measurement capability of the testbed. With such testing framework in place, application developers and service providers can better understand how radio channel conditions and network resources will influence the QoE and what specific improvements of the elements in the chain will bring big impact on the user perceived QoE.

The rest of the paper is organized as follows. A state-of-the-art review of QoE assessment is presented in Section 2. Use cases and network scenarios are defined in Section 3. The measurement methodology is detailed in Section 4. The calibration of the testbed is described in Section 5. The QoE measurement setup and a case study of two popular use cases are presented in Section 6. Finally, conclusions are drawn in Section 7.

## 2. Related work

QoE modeling and assessment is an important and active research area. A survey on QoE assessment approaches including subjective assessment, objective assessment, and hybrid assessment is reviewed in [8]. In the literature, most of the studies have been focused on the QoE assessment for video streaming. Song et al. [9] proposed an acceptability-based QoE model based on the results of comprehensive user studies on subjective quality acceptance assessments. Statistical nonlinear regression analysis is applied to build the model with a set of influencing factors such as encoding parameters, video content characteristics, and device display resolution. A Continuous QoE evaluation framework for video streaming subjected to rate adaptation and rebuffering distortions is proposed in [10], which consists of learning based and parametrized QoE evaluation methods. A survey on application-oriented, machine learning based QoE prediction models for multimedia services with focus on extended reality and video gaming is outlined in [11]. Barman et al. [12] presented a comprehensive overview of recent works of QoE modeling for HTTP adaptive streaming, with focus on the QoE models, influence factors, subjective test methodologies, as well as open challenges. Vučić et al. [13] investigated the key aspects in terms of different video encoding parameters influencing subjective QoE scores for multiparty audiovisual calls. Results indicate that it is better to provide constant lower objective video quality than to switch between higher and lower qualities in certain cases.

Besides the research from the academia, there are several ITU-T recommendations which provide an overview of algorithmic models for audio, video and audio-visual quality estimation based on IP headers and bit stream. Input information on more general aspects of the stream, such as video resolution, which may not be available in the packet header information, is provided to the algorithm in the out-of-band model, e.g., in the form of stream-specific side information. ITU-T Recommendation P.1201 specifies the model algorithm for low and high resolution [14]. The model algorithms are reference-free, which means that they are operated by analyzing the packet data captured in the network. The main difference with ITU-T P.1202 is that ITU-T P.1201 models provide audio, video and audio-visual quality estimates, while ITU-T P.1202 models only provide video quality estimates [15]. On the other hand, ITU-T P.1203 describes the algorithms of the models for monitoring the E2E quality of the media session for TCP-type video transmission [16]. The input used by the models consists of the information obtained by pre-inspection of the stream. Four different levels of inspection are included, resulting in models of different complexity of both the input information and the model algorithms, which are referred to as “modes of operation”. Finally, ITU-T P.1204 describes the model algorithms for video quality monitoring for streaming using reliable transport [17]. Input information is based on bit stream, pixel and hybrid (using both bit stream and pixel information) over reliable transport for resolutions up to 4 K. Recommendations ITU-T P.1203 and ITU-T P.1204 address two application areas: large screen presentation, as in the case of fixed network video transmission, and mobile transmission on handheld devices, such as smartphones. As a result, each of the algorithms in the recommendation models provides individual estimates of audio, video and audio-visual quality in terms of MOS scale of the Absolute Category Rating (ACR).

Although there are many proposals for QoE modeling and assessment, they are mostly oriented to a specific service, for example voice or video streaming. This paper proposes a framework that can generalize the QoE computation for any applications or services, with the aim to produce a single quality score in each measured domain which could eventually be used to certify applications based on common consensus.

Besides the different QoE assessment approaches mentioned above, various types of testbeds, ranging from simulated to emulated testbeds, have also been proposed to obtain QoE metrics, to extract a QoE model, or to assess the correctness of the QoE model. QoE-Lab [18] is a multi-purpose heterogeneous testbed that supports a variety of networking conditions to evaluate QoE of different applications. It supports background traffic generation, network emulation, seamless mobility between different access technologies, and interconnection between experimental setup and the Internet. The main limitation of this testbed is that it does not support wireless access technologies such as LTE. QoE Doctor [19] is a tool that supports accurate, systematic, and repeatable measurements and analysis of mobile app QoE, which uses user interface (UI) automation techniques to replay QoE-related user behavior, and measures the user perceived latency directly from UI changes. The main drawback of this approach is that most metrics are based on detecting specific UI changes, requiring module modification for each specific app under test. A testbed for evaluating the QoE of 3D video streaming service over LTE is described in [20], which consists of a streaming server, an internet protocol-level mobile network emulator, and a streaming client. A simulation campaign has to be carried out in a system-level LTE simulator to obtain the network emulator configuration parameters for different network conditions, which is a time consuming task.

From the literature it is evident that the proper configuration of a vast number of parameters to emulate different network conditions is of great importance for QoE evaluation, which requires a deep understanding of network details. This paper aims to simplify this procedure by defining a set of pre-configured network scenarios that an application or a service could encounter in realistic situations.

**Table 1**
Description of the TRIANGLE use cases.

| Identifier | Use case | Description |
|---|---|---|
| VR | Virtual Reality | Virtual reality content distribution |
| GA | Gaming | Connected online games |
| AR | Augmented Reality | Distribution of virtual content based on the perceived camera image |
| CS | Content Distribution Streaming Services | Content streaming of e.g. music or videos |
| LS | Live Streaming Services | Live streaming from the devices |
| SN | Social Networking | Sharing of text posts, images, or offline video |
| HS | High Speed Internet | Web browsing |
| PM | Patient Monitoring | Online monitor and collection of patient vitals, remote consultations |
| ES | Emergency Services | Connected emergencies of public services (ambulances, police, etc.) |
| SM | Smart Metering | Collection of sensor data (e.g. pluviometer, air quality) |
| SG | Smart Grids | Sensing and actuation on power grids |
| CV | Connected Vehicles | V2X systems, distribution of critical data, including self-driving |

## 3. Use cases and network scenarios

The TRIANGLE project aims to define a framework to test and benchmark the E2E QoE performance of applications, services and devices on various domains under realistic deployment conditions. Although QoE is perceived as a subjective metric, it is a key measure for the user's satisfaction with a specific service. The QoE evaluation is tightly associated with the type of use case. For example, the indicators such as time to login, time to post or to download content are relevant Key Performance Indicators (KPIs) for social network use case. While response time, video resolution and frames per second are relevant KPIs for virtual reality use case. Therefore one of the very first steps to define a test and certification framework is to identify the proper use cases and relevant network scenarios. This approach was initially defined by NGMN [21] and was adopted by 5G-PPP Association [22], where the use cases and the network scenarios were tightly coupled. In our work, we decoupled the use cases from the network scenarios for broader scope of investigation.

### 3.1. Use cases

A use case defines the type of application, service, or vertical that is using the network service. The use cases identified in TRIANGLE have been abstracted in order to ensure a homogeneous certification scheme, and in TRIANGLE's view, they represent the vast majority of the applications and services that are expected to travel through today's 4G and the coming 5G networks. This is essential in order to better understand which part of the QoE evaluation should be considered. The full list of use cases identified in TRIANGLE is outlined in Table 1, which includes services normally accessed via mobile devices and those targeted for Internet of Things (IoT) applications. The specification of each use case was defined in [3].

### 3.2. Network scenarios

The other essential component for the testing methodology is the network scenarios. The network scenario defines where the service is being used and how the operational condition is, which has a great impact on the QoE performance evaluation. It allows replication of real-life conditions for individual user in a lab-controlled environment. One major difference as compared to the existing work in the field is that the scenarios defined in our work do not look at the aggregated values from the network perspective but rather to the individual user, i.e., the scenarios define the exact conditions that the user will experience.

**Table 2**
List of the TRIANGLE network scenarios.

| Scenario | Channel model | Channel correlation | Doppler shift |
|---|---|---|---|
| Urban | | | |
| Urban-Office, indoor conditions (UR-OF) | EPA | Medium | 5 Hz |
| Urban-Pedestrian, outdoor slow mobility (UR-PE) | EPA | Low - medium | 5 Hz |
| Urban-Driving-Average driving (UR-DN) | EVA | Medium | 50–75 Hz |
| Urban-Driving-Traffic jam (UR-DT) | EVA | Medium | 25–150 Hz |
| Urban-Driving-Emergency driving (UR-DE) | EVA | Medium | 150 Hz |
| Urban-Internet Cafè, Busy Hours (UR-IB) | EPA | Low - medium | 5–20 Hz |
| Urban-Internet Cafè, Off-Peak (UR-IO) | EPA | Low - medium | 5–20 Hz |
| Sub-Urban | | | |
| Suburban-Festival (SU-FE) | ETU | Medium | 5 Hz |
| Suburban-Stadium (SU-ST) | ETU | Medium | 5 Hz |
| Suburban-Shopping Mall, Busy Hours (SU-SB) | ETU | High | 5 Hz |
| Suburban-Shopping Mall, Off-Peak (SU-SO) | ETU | High | 5–70 Hz |

Each network scenario wraps the full specification of the detailed test and configuration parameters that an application or a service could encounter in realistic situations, ranging from the radio channel conditions (e.g., received signal power, channel model, Doppler frequency) to the network conditions (e.g., number of active users, scheduling mechanism, available frequency/time resources). The proper configuration of all the detailed parameters is of great importance for testing and performance bench-marking, which requires multi-disciplinary knowledge, such as physical layer transmission parameters, protocols, radio propagation conditions, etc. The objective of defining various network scenarios is to simplify the testing procedures by avoiding the complexity of configuration details. A user can simply choose one or more scenarios relevant for his use case from a list, instead of defining hundreds of cellular network parameters, which requires a full understanding of the testbed details and is very time consuming.

The network scenarios developed in TRIANGLE are listed in Table 2. Each network scenario is composed of several sub-scenarios, capturing the different characteristics of radio channel and network condition variations within each sub-scenario. For instance, during the lab emulation of the Suburban-Festival network scenario, the amount of available frequency and time domain resources to the user will be changing dynamically, to emulate downlink-starved situations (which can be experienced at a festival entrance where many attendees download their e-tickets) as well as uplink-starved situations (which could occur when a popular band is playing and the crowd is live-streaming the performance on social media). In addition to the emulation of realistic network conditions, the testbed also includes a channel emulator which can employ various 3GPP specified channel models (e.g., EPA, EVA, ETU) together with user mobility (Doppler spread) configuration to fully represent radio channel conditions in a lab-controlled environment. The users can also define their own scenarios based on the use case and the KPIs of interest. Detailed configurations of each network scenario defined in TRIANGLE can be found in Annex B from Table 44 to Table 52 in [23].

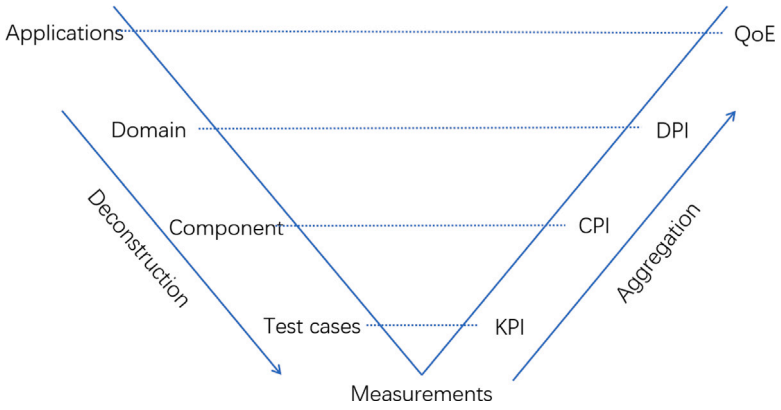



**Fig. 1.** Measurement methodology.

## 4. Testing methodology

The considered use cases in this project have been defined in Section 3.1. Each use case is typically associated with one or several domains for QoE evaluation. Domain is a categorization of relevant characteristics related to a specific subject, such as user experience, resource usage, energy consumption, etc. Domains are orthogonal to each other. One major challenge here is that domains normally cannot be evaluated directly due to their complexity, e.g., how do you test user experience of a mobile application? In order to make this evaluation feasible, a systematic approach that can transform the domain evaluation into a number of actionable tests running on the testbed and generalize the QoE computation for any domain is needed. Toward that end, we propose a methodology that consists of two processes: deconstruction and aggregation. Then KPI normalization is performed for further QoE computations.

### 4.1. Deconstruction and aggregation

The deconstruction process is responsible for decomposing a domain into multiple basic elements that can be physically measured. As shown in Fig. 1, each application can be measured with several domains. Each domain is deconstructed into several components, which characterize one specific aspect of the domain performance. The components are the key composing elements required to evaluate the performance of the associated domain. Each component is associated with one or more test cases. A test case describes the objective of the test and what needs to be tested. It specifies the initial conditions for the tests, identifies the required inputs and the expected outputs, and provides precise procedures for test case execution. Each test case intends to test an individual feature, aspect, or behavior of the application under test. For example, for the Content Streaming (CS) use case, a test case of "Non Interactive Playback" in the user experience domain will test if a device or an application can play a media file properly. Specifically, the test will perform several measurements related to the "Non Interactive Playback" test case, including the time to load the first media, the probability that a successfully started video stream is ended by a cause other than the user intention, the elapsed duration of content stalls while playing the media, and video resolution. This example test case might be only applicable for CS use case. The test cases defined in this project are written following the same pattern as defined in Global Certification Forum (GCF) for easy adoption by the certification entities if accepted. In addition, the test case describes the measurements that need to be executed by the testbed. The measurements refer to the basic elements that can be physically measured by the testbed. Examples of measurements that can be provided by the testbed are current, voltage, delay, jitter, throughput, block error rate (BLER), exchanged messages, logs (at the different layers), etc.

On the other hand, the aggregation process takes these basic evaluations and aggregates them into more abstract scoring and repeats the process until a score for the domain is achieved. The methodology behind the aggregation process is depicted in Fig. 2, which describes how the measurements are processed using data-driven methods for different test cases. The execution of the test case will generate a large amount of raw measurements data, from which one or more KPIs can be extracted through statistical analysis. KPIs are the performance values associated with a specific feature. Several measurements can be combined to extract different KPIs. As individual KPIs are measured in different dimensions and scales, they are normalized into a standard 1-to −5 scale, referred to as synthetic MOS, for further QoE computations. This process has been adopted by NGMN and other previous works [21,24]. The normalization of KPIs is a critical step in QoE computation framework and will be explained in more details in Section 4.2. Once the measured KPIs have been converted into synthetic MOS, they are used for homogeneous comparison and further QoE computations. Specifically, synthetic MOS are first aggregated to obtain the Component Performance Indicators (CPIs), which are the performance values of different components deconstructed from each domain during the deconstruction process. Then the Domain Performance Indicator (DPI), which is the domain score, is calculated by aggregating all the associated CPIs. Based on these DPIs, the certification or performance benchmarking of a specific application/service can be done.

The aggregation is performed using weight average over the measured KPIs, CPIs, and DPIs. The specific weights used in the aggregation depend on how important these KPIs/CPIs/DPIs are with respect to each other. The setting of the weights is quite subjective. The same weight is applied in this study. Of course, different weights can be assigned during the aggregation process based on the importance of the tested features for the application under test. The aggregation is also performed across different scenarios using weight average. The weight for each scenario should follow the relevance that the application could encounter in realistic situations.

### 4.2. KPI normalization

As different KPIs are measured in different dimensions and scales, the normalization of KPIs into a standard 1-to −5 scale is necessary for homogeneous comparison and aggregation. The conversion is based on the normalization function and normalization parameters. The normalization function can perform linear or logarithmic interpolation on the original KPI, or other interpolation functions with respect to the characteristics of the measured KPI.

For linear interpolation, the synthetic MOS value in the range [1-to −5] can be calculated as:

$$v'_{\mathrm{kpi}} = \begin{cases} \mathrm{KPI}_{\min}, & \text{if } v_{\mathrm{kpi}} < \mathrm{KPI}_{\min} \\ \mathrm{KPI}_{\max}, & \text{if } v_{\mathrm{kpi}} > \mathrm{KPI}_{\max} \\ v_{\mathrm{kpi}}, & \text{if } \mathrm{KPI}_{\min} \le v_{\mathrm{kpi}} \le \mathrm{KPI}_{\max} \end{cases} \quad (1)$$

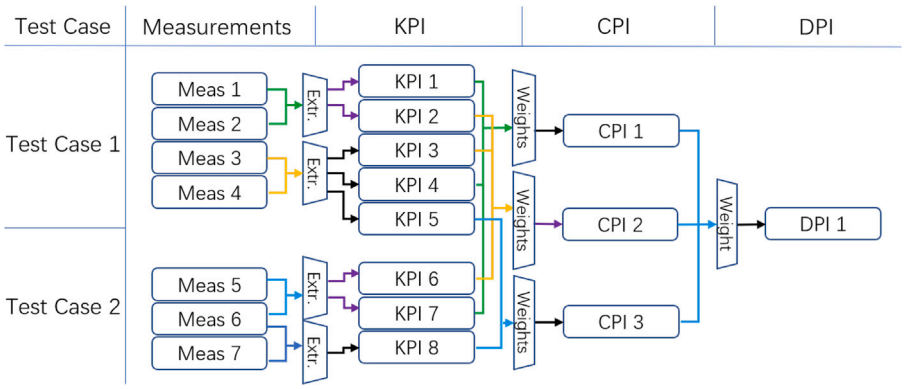



**Fig. 2.** From measurements to domain performance indicator.

$$v_{\text{mos}}^{\text{linear}} = \frac{v'_{\text{kpi}} - \text{KPI}_{\text{min}}}{\text{KPI}_{\text{max}} - \text{KPI}_{\text{min}}} \cdot (5.0 - 1.0) + 1.0 \tag{2}$$

where $v_{\text{kpi}}$, $v'_{\text{kpi}}$ and $v_{\text{mos}}^{\text{linear}}$ are the measured KPI, truncated KPI, and linearly normalized KPI, respectively. $\text{KPI}_{\text{min}}$ and $\text{KPI}_{\text{max}}$ are the normalization parameters that define the worst and the best reference benchmarking values of a specific KPI, respectively. If the measured KPI $v_{\text{kpi}}$ falls out of the range between $\text{KPI}_{\text{min}}$ and $\text{KPI}_{\text{max}}$, the truncated value $v'_{\text{kpi}}$ will be set to either $\text{KPI}_{\text{min}}$ (if it is worse) or $\text{KPI}_{\text{max}}$ (if it is better). It converts a measured KPI into a synthetic MOS by applying a simple linear interpolation between the worst and the best expected values from a reference case.

The logarithmic interpolation is inspired by the opinion model recommended by the ITU-T for simple web search task [25]. For logarithmic interpolation, the synthetic MOS in the range [1-to −5] can be calculated as:

$$v_{\text{mos}}^{\text{log}} = \frac{\ln(v'_{\text{kpi}}) - \ln(\text{KPI}_{\text{min}})}{\ln(a \cdot \text{KPI}_{\text{min}} + b) - \ln(\text{KPI}_{\text{min}})} \cdot (5.0 - 1.0) + 1.0 \tag{3}$$

where $a$ and $b$ correspond to the simple web search task case with default values 0.003 and 0.12, respectively [25,26]. The values for parameters $a$ and $b$ shall be updated according to the interpolation requirement if through subjective experimentation other values are found to be better adjustments for specific services.

Besides the above mentioned linear and logarithmic interpolations, other linear or non-linear interpolation functions can be applied to best map the measured KPI into a standard scale. Furthermore, machine learning is a powerful tool for data fitting.

Due to the dynamics introduced in the network scenarios, a test campaign is executed for each test case with a large number of iterations (e.g., > 20) in each supported scenario in order to obtain statistical consistency and get sufficient number of samples per KPI per scenario. The convergence analysis with respect to the number of required iterations will be introduced in more details in Section 5.2. Then the normalization function and normalization parameters can be derived through statistical analysis of the measured KPI samples, e.g., mean, standard deviation, cumulative distribution function (CDF), ratio, etc. In our study, different combinations of normalization functions and normalization parameters have been experimented with the objective that the converted synthetic MOS should capture the behavior of the measured KPI in different scenarios and should match well with human perception. Once all the measured KPIs are translated into synthetic MOS, they can be averaged with suitable weights for further QoE calculations.

### 4.3. Measurement methodology

The overall testing and certification process using the TRIANGLE testbed is illustrated in Fig. 3, which can be decomposed into two basic blocks, i.e., the measurement block and the ETL process block. The measurement block is responsible for performing measurements of various KPIs related to specific use cases and test cases, while the ETL process is responsible for KPI extraction, normalization, and aggregation from measurements towards the final QoE score for each considered domain. The proposed QoE evaluation framework consists of the following steps:

- **Step (1) - Run Tests:** the testbed automatically loads and runs the required test cases related to a specific domain. The deconstruction process is performed in this step to decompose a domain into multiple relevant test cases. Each test case has to run a number of iterations. This step is completed when the testbed has collected all the measurements required. The raw measurement data are stored in a database for later processing.
- **Step (2) - Run Analytics:** the KPIs can be extracted through statistical analysis from the raw measurement data. For example, the KPI "time to load the first media" in the "Non Interactive Playback" test case can be obtained by calculating the mean of the measured values across all iterations.
- **Step (3) - Compute Metrics:** the measured KPIs from the previous step are then converted into the synthetic MOS for further QoE calculations. The aggregation process depicted in Fig. 2 is performed in this step to calculate the CPI and DPI for each domain.
- **Step (4) - Output Score:** once the DPIs for all the tested domains have been calculated, a final single mark that illustrates the overall performance of the application/service/device under test can be generated using weight average across the domains.

## 5. Testbed calibration

An important pre-requisite before executing any test campaign is to calibrate the testbed to ensure that it can provide consistent and repeatable measurement results. The testbed has to be calibrated with respect to the measurements as well as the ETL process. The calibration of the measurements includes calibrating the individual instrument independently as well as characterizing the overall performance of the testbed. Apart from that, statistical analysis of the measurement results is also needed to ensure that the measurements converge towards a stable test result. The calibration of the ETL process is to ensure that the conversion from the measured KPIs into the synthetic MOS values and the computation of the QoE score in each measured domain are correct, i.e., it matches with the actual human perception. The detailed procedures for testbed calibration is described in [27].

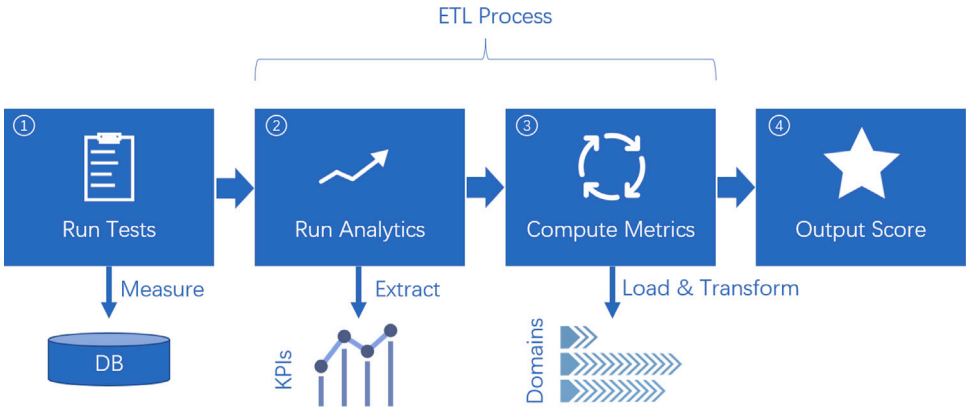



**Fig. 3.** From measurements to QoE score.

### 5.1. Measurement calibration

The testbed is comprised of various measurement instruments (e.g., power analyser, base station emulator, commercial phones, etc.), software components (e.g., data performance tool, test orchestration tool, mobile test automation tool, etc.), and other hardware components. It needs to be calibrated and proper compensations for certain measurements have to be adjusted against the baseline before executing any test campaign. The necessary procedures needed for the measurement calibration include:

- **Calibration of the individual measurement instrument:** The calibration of the individual measurement instrument shall be done according to standard procedures by calibration laboratories with recognized traceability.
- **Characterization of the testbed:** The characterization of the overall performance of the testbed includes latency and throughput characterization, which are the two most important parameters to establish the baseline of the testbed.
- **Compensation of cabling loss:** The cable that connects the device and the testbed may introduce additional attenuation. Its impact on the measurement results has to be characterized for proper compensation. It shall be performed every time a new cable or a new device is connected to the testbed, and shall be performed on a periodic basis to compensate for the drift of the cable characteristics over time.

The outcome of the measurement calibration is a testbed calibration and compensation table, which can be served as the baseline reference and will be used as inputs to the ETL process of QoE computations. An example of a testbed calibration and compensation table generated from the TRIANGLE testbed is shown in Table 3.

Whenever an event which could potentially cause changes in the environmental settings of the testbed occurs or a new element has been replaced/added into the testbed, the procedure of measurement calibration shall be repeated accordingly.

**Table 3**
Testbed calibration and compensation table from TRIANGLE testbed.

| Parameter | Value |
|---|---|
| Testbed Name | TRIANGLE |
| Internal Latency (RTT) | 25 ms to server A |
| External Latency (RTT) | 36 ms to server B (8.8.8.8) |
| Max Throughput | 204.87 Mbps to server A (local)<br>92.87 Mbps to server B (remote) |
| RF Loss Compensation | −2.0499 dB on Band 1 (TXRX1)<br>−2.3500 dB on Band 1 (TXRX2)<br>−1.8500 dB on Band 3 (TXRX1)<br>−1.9500 dB on Band 3 (TXRX2)<br>−4.3700 dB on Band 7 (TXRX1)<br>−4.5500 dB on Band 7 (TXRX2)<br>−1.3500 dB on Band 20 (TXRX1)<br>−1.5500 dB on Band 20 (TXRX2) |

### 5.2. Convergence study

Another important factor affecting the measurement results is the dynamics introduced in the network scenario, which have been described in Section 3.2. The network scenarios define various properties of the network conditions (e.g., the number of active users, the amount of available time/frequency resources for scheduling, etc.) and the radio channel conditions (e.g., channel models, Doppler frequency, received signal power, etc.), which have great impact on the overall performance. Because of the statistical nature of the dynamics introduced in the network scenario, the measurement results may vary from one iteration to another. To reduce the measurement uncertainty, in each run of the test case the measurements are performed several times, assuming that the values are affected by a random process and the measurement error can be diminished by averaging the results over various iterations. Therefore, it is important to characterize the repeatability and consistency of the measurement results, i.e., does the testbed generate consistent results when the test is performed at different places and/or different time? Also, Statistical analysis is needed in order to understand whether the average is a good estimation of the measurement results and how the measurements converge towards a stable result, as well as to identify the number of test iterations needed to obtain a stable and converged result in each network scenario. The latter is especially important to strike a good balance between the accuracy of the measurements and the time required to achieve that accuracy.

To characterize the performance of different network scenarios, four traffic profiles have been tested, covering synthetic data (TCP and UDP streams), YouTube video streaming, and Spotify audio streaming. As mentioned in Section 3.2, each network scenario is composed of several sub-scenarios with different characteristics of radio and network conditions. The duration of each test iteration should be long enough to make sure that each test iteration properly covers all the sub-scenarios. As the longest network scenario lasts for 150 s, the minimum duration of each test iteration was set to be 180 s. The configured traffic profiles for testbed calibration are listed in Table 4.

To study the consistency and repeatability performance of the testbed, two replicas of the testbed, namely Testbed A and Testbed B, have been built, located in Spain and Denmark, respectively. The throughput and data convergence performance with TCP x1 stream traffic in urban and suburban scenarios are shown in Fig. 4 and Fig. 5, respectively. The $x$-axis is the number of iterations. The $y$-axis in the convergence subplot indicates the absolute difference between

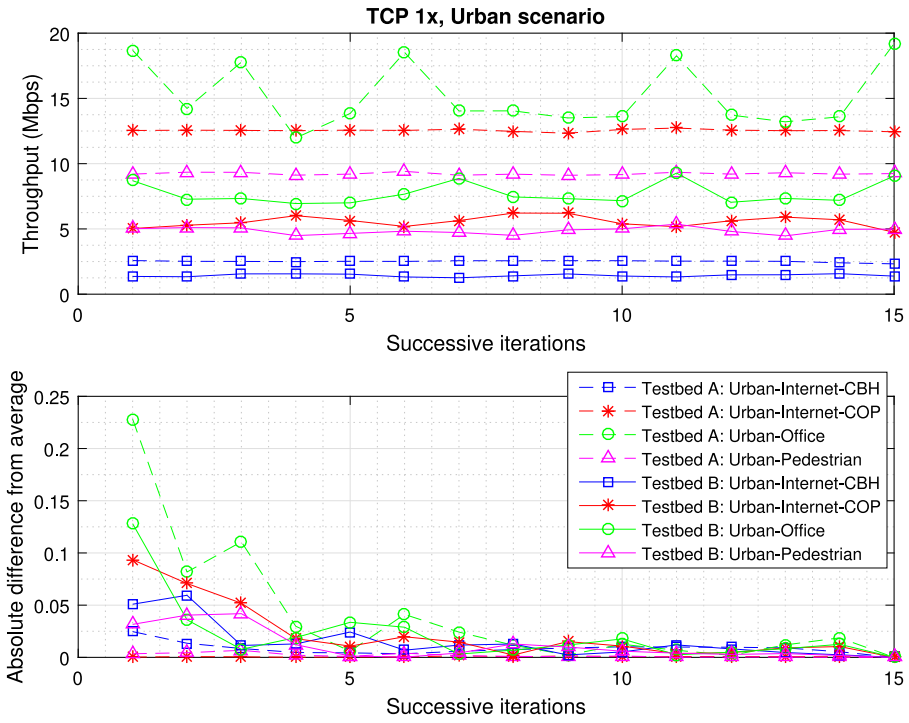



**Fig. 4.** Convergence performance with TCP traffic in Urban scenarios.

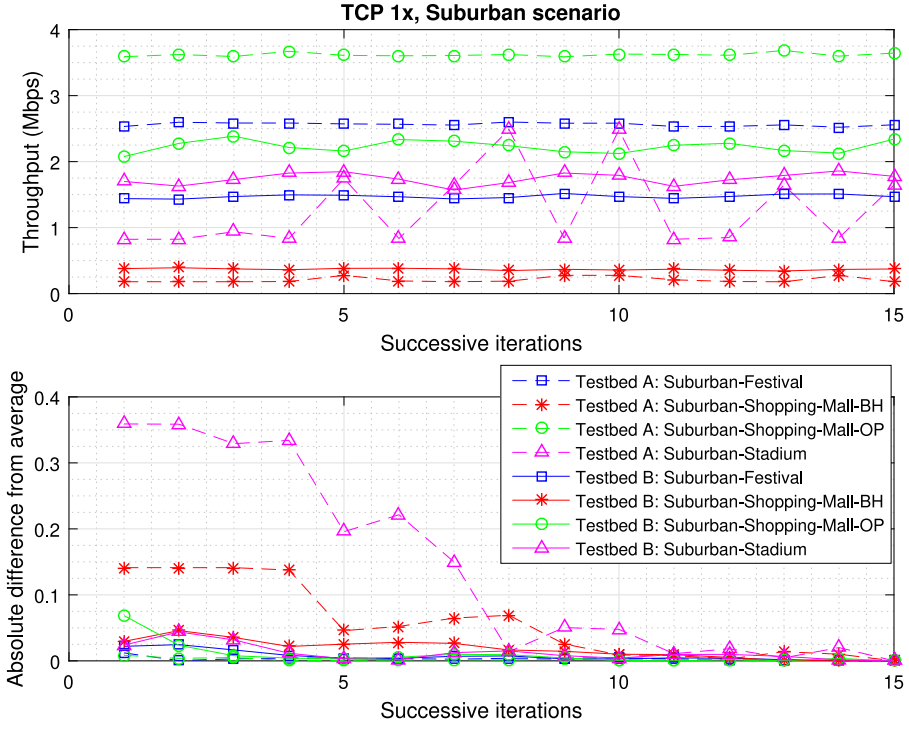



**Fig. 5.** Convergence performance with TCP traffic in Suburban scenarios.

the total average (15 iterations) and the average after X number of iterations. From the throughput subplot, it is clearly shown that there are fluctuations from one test iteration to another due to the changing conditions of the radio channel and the available network resources in the associated sub-scenarios. The convergence subplot indicates that the number of test iterations needed for the average to converge depends on the configuration of the network scenario. For scenarios with fast changing radio channel and network resources higher number of iterations are needed to achieve a stable result, e.g., within 5% variation. It is observed from the measurements that generally more than 10 iterations are required for TCP traffic. For UDP traffic, although not shown in the paper due to the limited space, quite stable results can be achieved with a few number of iterations if the traffic load is less than the channel capacity. However for UDP traffic load close to or higher than the channel capacity, a larger number of iterations (e.g., more than 15 iterations) are needed to average

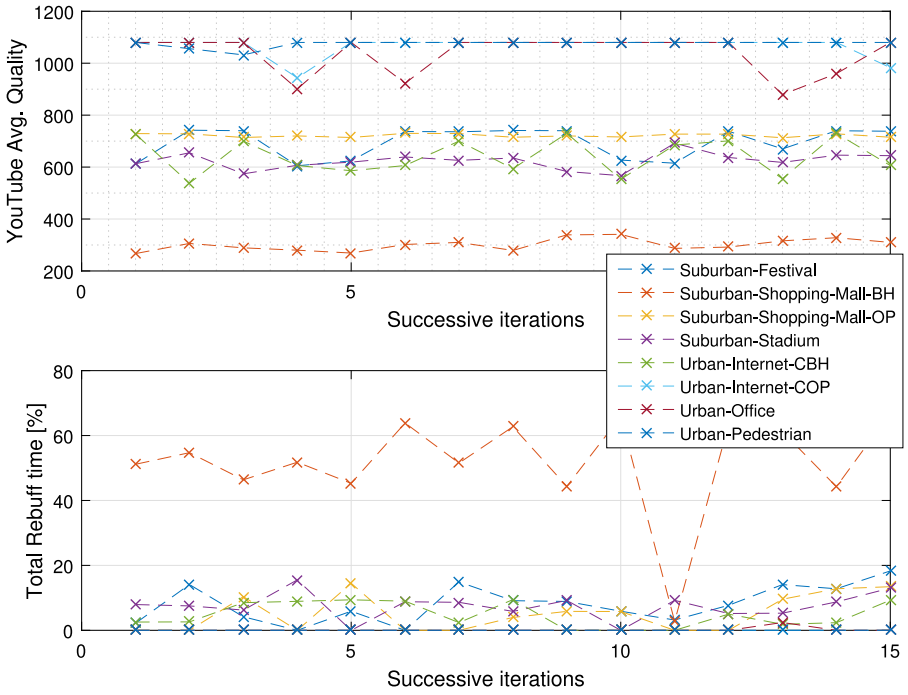



**Fig. 6.** Average video quality and total re-buffering time in different network scenarios.

**Table 4**
Configured traffic profile for testbed calibration.

| Traffic profile | Test duration | KPI |
|---|---|---|
| TCP x1 stream | 180 s | Throughput |
| TCP x4 streams | 180 s | Throughput |
| UDP @ 128 kbps | 180 s | Throughput, delay jitter, packetloss |
| UDP @ 2 Mbps | 180 s | Throughput, delay jitter, packetloss |
| YouTube<br>1080p @ 30fps<br>1440p @ 30fps<br>1080p @ 60fps | 300 s | Video resolution<br>Re-buffering time |
| Spotify<br>HQ Audio (192 kbps) | 300 s | Re-buffering time |

out the variations. It is worth mentioning that although the absolute throughput measured at Testbed A and Testbed B are different due to the use of different phones, the convergence performance from the two testbeds are comparable with each other.

Fig. 6 shows the total re-buffering time and the average video quality in terms of video resolution for YouTube video streaming under different network scenarios. It is observed that the performance of the total re-buffering time has more fluctuations than the average video quality and therefore is the main KPI that determines the number of iterations needed for the convergence of the average. Again, the specific number of iterations is scenario dependent. Similar behavior is also observed for the test case of Spotify audio streaming.

From the above study, it is shown that the number of iterations needed for the averaged results to converge depends both on the specific network scenario and the configured traffic profile. Even with the same traffic profile, it is necessary to configure different number of iterations with respect to the configured network scenario. Nonetheless, a key finding is that the system does converge with feasible number of iterations, i.e., in most cases only 15 iterations are needed for the averaged measurement results to converge.

### 5.3. ETL process calibration

The above mentioned calibration steps, i.e., the calibration of individual instrument, the characterization of testbed performance, and the convergence analysis, can be categorized into the calibration of the measurements. The next step following the measurements is the ETL process which aims to convert the measured KPIs into a subjective QoE score for each measured domain.

The calibration of the ETL process consists of the calibration of the conversion from the measured KPIs into the synthetic MOS values and the calibration of QoE computation. As the QoE assessment is quite subjective and there is no standardized calibration procedure for this process, human panel experiments are considered to be a proper approach to calibrate and verify if the testbed generated QoE score matches well with the human experienced QoE.

As a case study, we selected ExoPlayer, which is an open source media player originally developed by Google with support for DASH adaptive playbacks, as the application under test. This application falls into the use case of content streaming. There are 8 test cases defined in the TRIANGLE project for testing content distribution streaming service [23]. Due to the lack of space, here we only focus on a single test case of "Non-Interactive Playback", which is related to the application user experience (AUE) domain. It is worth mentioning that the testbed has the capability of performing various measurements in multiple domains. The configured test case for ExoPlayer and the measured KPIs are summarized in Table 5.

The methodology for the human panel calibration is that a set of videos have been reproduced under various network scenarios with the same testing conditions as in the testbed. The reproduced videos are 2–3 min long with a random order and without the presence of a reference clip. A group of 18 people with diverse ages and genders are invited to watch the reproduced videos in a random order. The number of people composing the human panel with respect to the gender and ages is shown in Fig. 7. A score sheet was given to each experimenter to set ratings with respect to the measured KPIs as well as the overall QoE score for each reproduced video, in the form of synthetic MOS value in a standard 1-to −5 scale, ranging from the lowest to the highest. The collected human panel experiment results for each

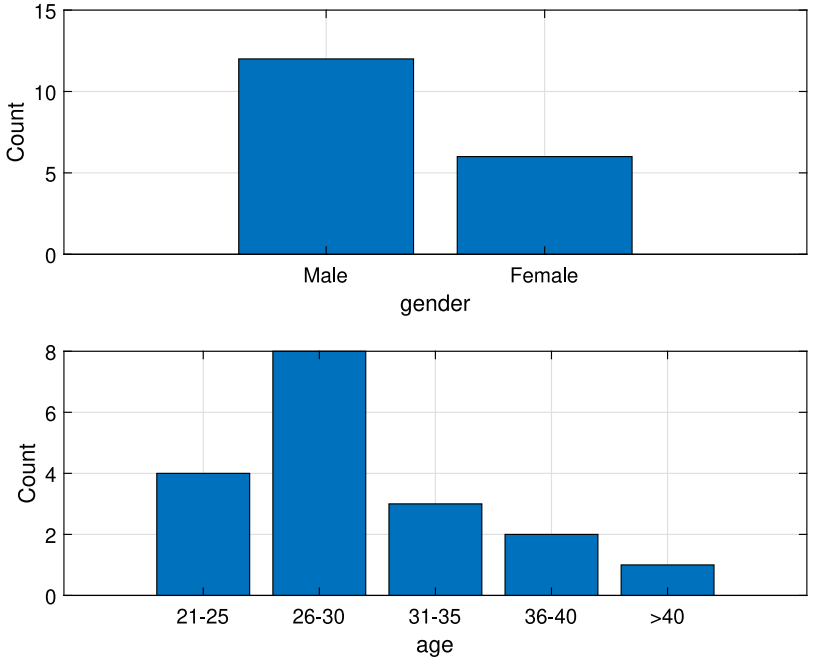



**Fig. 7.** Gender and age distribution of the human panel.

**Table 5**
Test configurations for ExoPlayer.

| | |
|---|---|
| Application under test | ExoPlayer |
| Reference test phone | Samsung Galaxy S7 |
| Use case | Content streaming |
| Test case | Non-Interactive Playback |
| Domain | Application user experience (AUE) |
| Measured KPIs in the AUE domain | • Video Start: The time elapsed since the user clicks play button until the media starts.<br>• Video Resolution: Used video resolution.<br>• Video End: Probability that a successfully started stream can be finished successfully.<br>• Smoothness: The duration of content stalls divided by total duration. |
| Network scenarios | All supported scenarios (11 in total) [23] |

measured KPI in each network scenario are averaged over the number of experimenters. Then the averaged results from the human panel are compared with the results generated from the testbed to calibrate the ETL process, i.e., the proper configurations of normalization function and normalization parameters for each measured KPI.

For a specific KPI $k$, the calibration is based on an objective function which is defined as to minimize the mean squared error (MSE) between the human experienced synthetic MOS value and the testbed generated synthetic MOS value over all measured scenarios, defined as:

$$\min_{f_k, \mathrm{KPI}^k_{\min}, \mathrm{KPI}^k_{\max}} \frac{1}{S} \sum_{s \in \mathrm{Scenarios}} \left(\theta_{k,s} - \mu_{k,s}\right)^2 \tag{4}$$

$$\mu_{k,s} = f_k(\lambda_{k,s}, \mathrm{KPI}^k_{\min}, \mathrm{KPI}^k_{\max}) \tag{5}$$

where $S$ denote the cardinality of set Scenarios, $f_k()$ and $\mathrm{KPI}^k_{\min}$, $\mathrm{KPI}^k_{\max}$ denote the normalization function and the worst and the best reference benchmarking values for a specific KPI $k$, respectively. As mentioned in Section 4.2, the normalization function can be linear/logarithmic interpolation, or other functions depending on the characteristics of the measured KPI. $\lambda_{k,s}$, $\theta_{k,s}$ and $\mu_{k,s}$ denote the measured KPI, the human experienced synthetic MOS, and the testbed generated synthetic MOS for KPI $k$ under network scenario $s$, respectively.

**Table 6**
Calibrated ETL configurations for KPI normalization for ExoPlayer.

| KPI | Normalization function | KPI_min | KPI_max |
|---|---|---|---|
| Video Start | Logarithmic | 10 s | 0.1 s |
| Video Resolution | Step | 144p | 1080p |
| Video End | Linear | 50% | 100% |
| Smoothness | Linear | 20% | 0 |

A comparison between the human experienced score and the testbed generated score before/after calibration for the measured KPIs in the "Non-Interactive Playback" test case specified in Table 5 under different network scenarios is shown in Fig. 8. Due to lack of space, 4 out of 11 network scenarios, which are Urban Office (UR-OF), Suburban Festival (UR-FE), Urban Driving Traffic Jam (UR-DT) and Suburban Shopping Mall Busy Hours (SU-SB), are plotted in the figure, representing good, median and bad network scenarios. It is observed that although the score generated by the testbed before the ETL calibration is quite similar to the human experienced score, there are some differences between the two scores. By fine tuning the ETL process with respect to the normalization function and normalization parameters for each measure KPI, the QoE score generated by the calibrated testbed can better align with the curves obtained from the human panel. The calibrated ETL configurations for KPI normalization for ExoPlayer are summarized in Table 6.

Once the measured KPIs have been normalized into the synthetic MOS, the aggregation process described in Section 4.1 can be performed to calculate the DPI for each domain through a domain-specific function (e.g., weight average). For example in the considered "Non-Interactive Playback" test case, the synthetic MOS value for the AUE domain in each network scenario can be obtained by averaging over the four measured KPIs, as shown in Fig. 9(a). Based on that, a single value representing the QoE score for the AUE domain, i.e., DPI, can be obtained by averaging the domain synthetic MOS values over the tested network scenarios, as shown in Fig. 9(b). Again, it can be seen that the testbed generated score matches quite well with human perception after the ETL process calibration. Specifically, the accuracy of the QoE computation from the testbed as compared to the human panel experiments can be improved from 87% to 97% after the calibration, in the AUE domain of the "Non-Interactive Playback" test case.

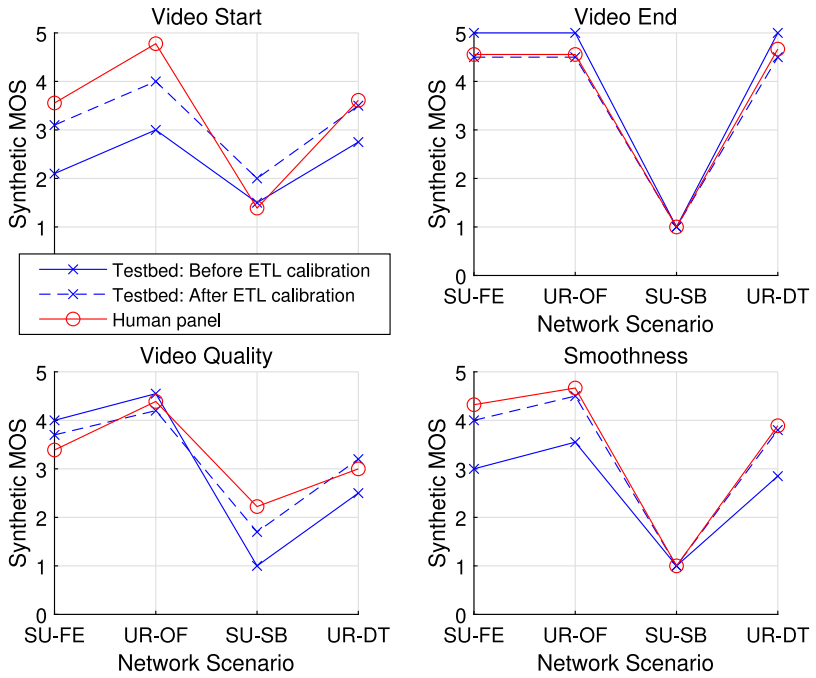



**Fig. 8.** Comparison between the testbed generated QoE score (with and w/o ETL calibration) and the human experienced score for different KPIs.

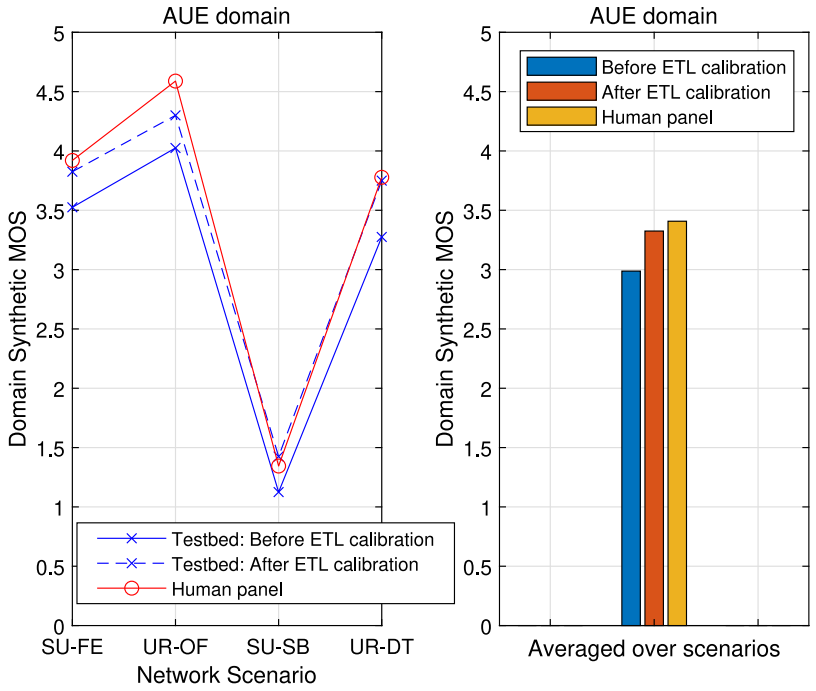



**Fig. 9.** (a) AUE domain synthetic MOS score for specific network scenarios. (b) QoE score for the AUE domain.

It is worth mentioning that the human panel experiments are performed on desktop screens in a silent and shadow office without much environmental noise, e.g., no strong sunlight reflection on the screen, limited background noise, the distance between the screen and all users was maintained, in order to diminish the negative impact that may affect the QoE scoring. In scenarios where the environmental noise is not neglectable which may affect the human perception of QoE, a scaling factor ranging from 0 to 1 has to be multiplied with the testbed generated QoE score to match with human perception. The choice of the scaling factor depends on the severity of the environmental noise.

## 6. Measurements

### 6.1. Measurement setup

The measurement setup is depicted in Fig. 10. The radio access emulator is provided by Keysight E7515 A UXM Wireless Test Set, which is a standard-compliant base station emulator including a channel emulator for the emulation of various complex radio channels. The mobile device under test is Sony Xperia F8331. The mobile device's antenna port is connected to the UXM via calibrated cabling. The purpose of using cabling instead of using Over-the-Air (OTA) connection is to enable

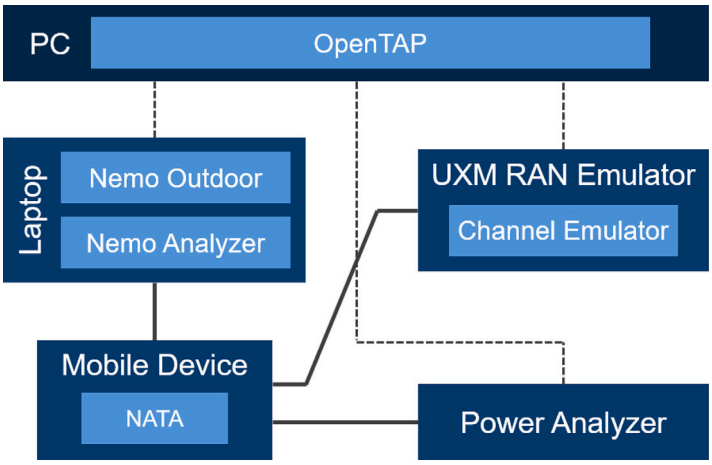



**Fig. 10.** Testbed setup.

accurate control over the radio channel conditions emulated by the UXM, thus allowing replication of real-life field conditions in a lab controlled environment and is repeatable. For the mobile devices and the applications under test, the UXM is perceived as a real network.

The mobile device is also connected to a laptop which has Nemo Outdoor and Nemo Analyze installed. Nemo Outdoor is a drive test tool which enables measuring QoE metrics for the services and applications the customers are using, including YouTube, Twitter, Facebook, Twitter, Dropbox, Viber, Instagram, etc. Various use cases and the associated test cases within each use case are implemented in Nemo Outdoor, specifying the actions and measurements that should be performed to compute the KPI of the feature under test. Nemo Analyze is a post-processing tool which is used to extract various KPIs from the measurement data produced by Nemo Outdoor. It also supports user defined template for further KPI calculations. The KPI normalization and aggregation process described in Sections 4.1 and 4.2 are performed in Nemo Analyze. Besides, the mobile device has Nemo Active Testing Application (NATA) tool installed, which supports real QoE measurements running on a smartphone and provides communication interface with Nemo Outdoor.

The testbed is controlled by Keysight's Test Automation Platform (TAP), which provides interfaces to the testbed components and orchestrates the behavior of different components. The configuration of the different components in the testbed is determined by the test case to run and the associated network scenarios. The network scenarios described in Section 3.2 are implemented in TAP, which controls the settings of the network scenario related parameters in UXM. To accurately analyze the power consumption during the experiments, the mobile device is directly powered by Keysight N6705B power analyzer.

### 6.2. QoE evaluation case study

To verify the measurement capabilities of the testbed and to benchmark the QoE performance of some well-known applications, we selected two use cases, namely content streaming and high-speed internet, as a case study. The domain of interest in this work is the user experience. For content streaming use case, the application under test is YouTube and ExoPlayer, and the selected test case is "Non-Interactive Playback", which corresponds to AUE/CS/001 in the TRIANGLE defined test cases [23]. For high-speed internet use case, the selected test cases are "Downloading files sequentially" and "Uploading files sequentially", which corresponds to AUE/HS/001 and AUE/HS/002 defined in [23]. The measured KPIs for the two considered use cases and the corresponding normalization function and parameter settings for each measured KPI are summarized in Table 7. The normalization function and parameters for the content streaming use case are obtained after the ETL process calibration with human panel (i.e., from Table 6). The normalization function and parameters for the high-speed internet use case are carefully selected to effectively map the raw measurements into synthetic MOS values based on experiments and human knowledge, but without human panel calibration due to the lack of time. The performance of the conversion from the measured KPIs into synthetic MOS values is analyzed in [27]. The ETL process calibration framework presented in Section 5.3 is valid for all use cases and can be used to fine tune the parameters for the KPI conversion. During the aggregation process, the same weight is applied.

Fig. 11 shows the normalized MOS for the measured KPIs for the content streaming use case under different network scenarios. As suggested in Section 5.2, 15 iterations are performed for each test case in each network scenario to average out the fluctuations of the measurement results caused by the variation of the radio channel conditions configured in each scenario. Due to the lack of space, only the most important 4 out of 6 KPIs for the "Non-Interactive Playback" test case are plotted. If we compare the KPI performance over different network scenarios, it is shown that the KPI performance varies from one network scenario to another. Generally the worst performance is often observed in the SU-SB scenario, while the best performance is often observed in the UR-OF scenario. That is because the SU-SB scenario is configured with more challenging radio channel conditions and limited network resources, while the UR-OF scenario is one of the scenarios with relatively good radio channel condition and sufficient network resources. Different channel conditions and available radio resources will lead to different RAN throughput, which further impacts the performance of the measured KPIs. Specifically, the performance of video resolution is sensitive to the network scenarios, while the performance of video end is not that sensitive to the network scenarios, except for some extreme scenarios (e.g., SU-SB). This observation is as expected because most of the content streaming applications can adapt its video resolution according to the experienced throughput so that the played video can be finished properly. Next if we compare the performance over the two content streaming applications, it is shown that YouTube has better performance than ExoPlayer in terms of smoothness and video resolution, while ExoPlayer shows advantage with respect to video start. The evaluation results from human panel experiments for ExoPlayer are also included in Fig. 11. It is shown that the MOS value generated by the calibrated testbed can match well with the scores obtained from the human panel. It is worth mentioning that during the human panel experiments, only 4 out of 11 network scenarios (i.e., UR-OF, UR-FE, UR-DT and SU-SB), representing good, median and bad network scenarios, are evaluated for the sake of not to overwhelm the human panel with too many scenarios.

By averaging over the measured KPIs, the QoE score in the user experience domain for the content streaming use case in each network

**Table 7**
The measured KPIs and the corresponding normalization for the two considered use cases of content streaming and high-speed internet.

| Use Case | KPI | Definition | Normalization | KPI_min | KPI_max |
|---|---|---|---|---|---|
| Content Streaming | App Access Time | Data connection establishment time | Logarithmic | 5 s | 0.1 s |
| Content Streaming | App Availability | Data server connection successful or not | Linear | 50% | 100% |
| Content Streaming | Video Start | Time to load first media frame | Logarithmic | 10 s | 0.1 s |
| Content Streaming | Video End | Probability of video stream can be finished successfully | Linear | 50% | 100% |
| Content Streaming | Video Resolution | Used video resolution | Step | 144p | 1080p |
| Content Streaming | Smoothness | The duration of content stalls divided by total duration | Linear | 20% | 0 |
| High-speed Internet | Access time | Data connection establishment time | Logarithmic | 5 s | 0.1 s |
| High-speed Internet | Accessibility | Data server connection successful or not | Linear | 0.5 | 1 |
| High-speed Internet | FTP DL throughput | Download throughput using FTP | Linear | 2 Mbps | 20 Mbps |
| High-speed Internet | FTP UL throughput | Upload throughput using FTP | Linear | 1 Mbps | 15 Mbps |
| High-speed Internet | HTTP DL throughput | Download throughput using HTTP | Linear | 2 Mbps | 20 Mbps |
| High-speed Internet | HTTP UL throughput | Upload throughput using HTTP | Linear | 1 Mbps | 15 Mbps |
| High-speed Internet | Transfer cut-off | Probability of file transfer can be finished successfully | Linear | 75% | 100% |

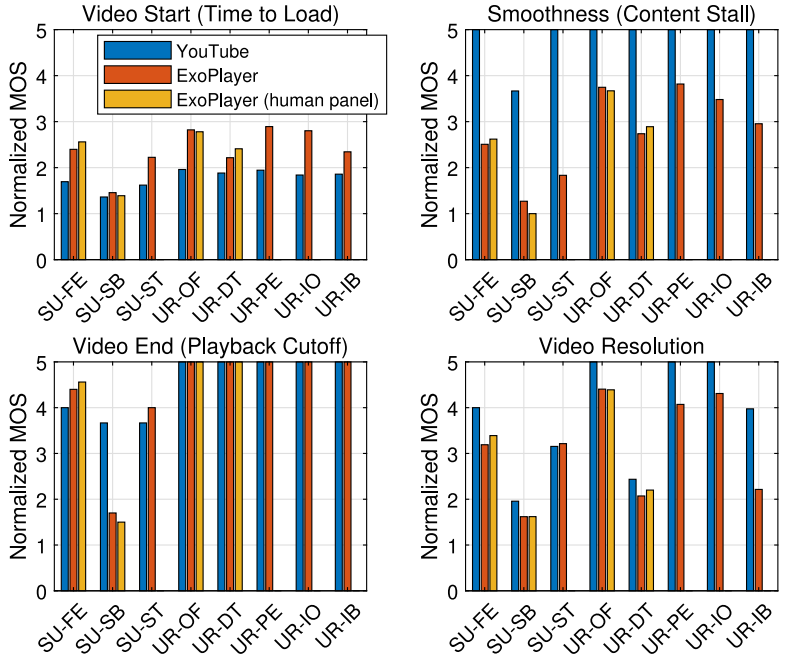



**Fig. 11.** Normalized MOS for the measured KPIs for the content streaming use case under different network scenarios.

scenario is depicted in Fig. 12. Again it can be seen that the QoE score varies across different network scenarios. If we compare the QoE performance of the two applications, YouTube shows slightly better performance than ExoPlayer, and the two applications exhibit similar performance variation pattern across the network scenarios. From a tester's point of view, it is useful to have a single value representing the QoE score for each domain. By averaging the QoE score over all network scenarios, the average QoE score in the user experience domain for the two tested content streaming applications can be obtained.

The same analysis has also been applied to the high-speed internet use case. Due to the lack of space, the normalized MOS for each measured KPI under different network scenarios is not shown in this paper. Similar behavior has been observed for the high-speed internet use case, as compared to the content streaming use case. Specifically, some KPIs are sensitive to the network scenarios (e.g., downlink/uplink throughput and transfer cut-off), while others are not (e.g., access time and accessibility). The QoE score averaged over the measured KPIs in the user experience domain for both content streaming and high-speed internet use cases under different network scenarios is depicted in Fig. 13. It is shown that the network scenarios have a great impact on the overall user experience. Thus when designing network scenarios, it is important to cover most of the testing conditions that

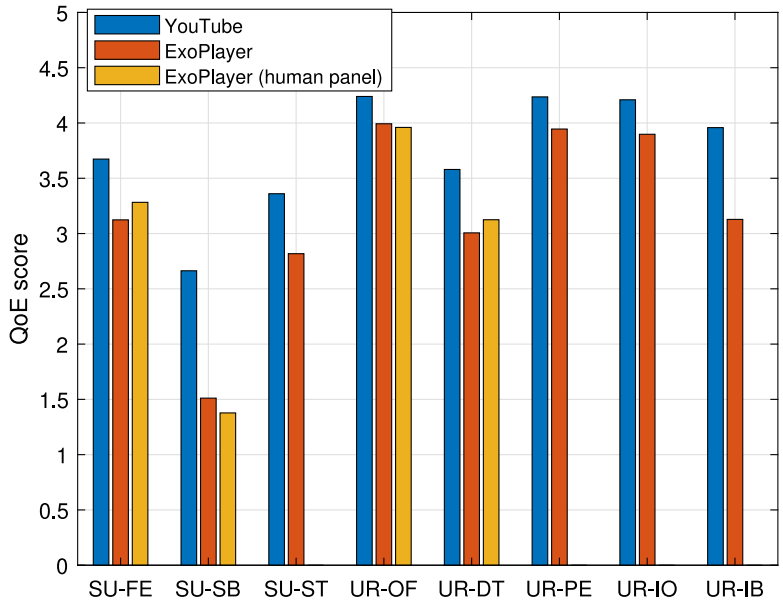



**Fig. 12.** QoE score in the user experience domain for the content streaming use case under different network scenarios.

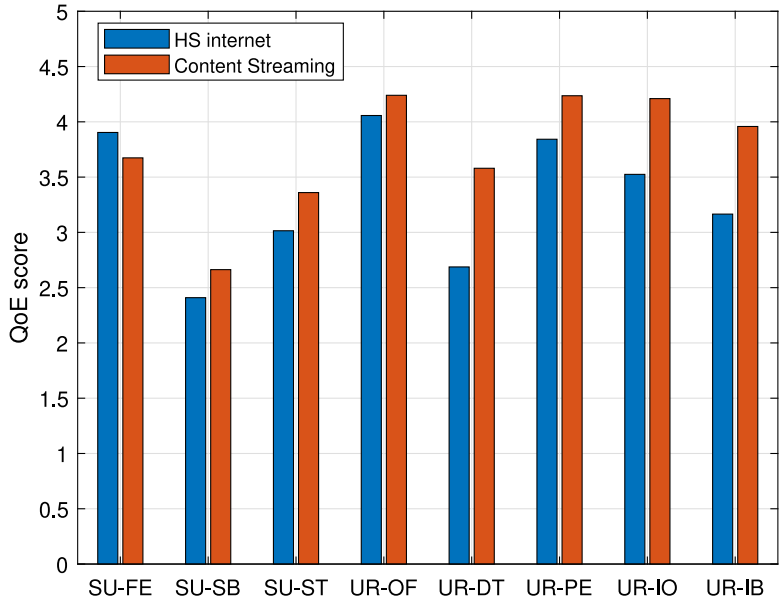



**Fig. 13.** QoE score in the user experience domain for both content streaming and high-speed internet use cases under different network scenarios.

the user/application will experience in real life. By comparing the QoE performance of the two use cases, it can be seen that although the absolute QoE value differs from one use case to the other due to different KPIs and normalization settings, similar performance pattern is observed for the two use cases, which validates the design of our testing methodology. The testbed is designed to have the capability of generalizing the QoE computation for any applications or services, and having full control over the testing environment including radio conditions as well as network resources, by identifying different network scenarios.

In this paper, we are focused on the QoE evaluation in the user experience domain, which only shows a part of the picture. In order to get the full picture of the application under test, additional domains such as resource usage, energy consumption, among others, should be evaluated in a similar way. By averaging the QoE score over multiple measured domains, a final QoE mark, the so-called TRIANGLE mark, can be obtained.

## 7. Conclusion

In this paper, we aim to design a 5G testing and benchmarking framework that can evaluate QoE for new mobile applications and devices in a realistic and controllable manner. The proper use cases and relevant network scenarios have been identified. The network scenarios allow replication of real-life conditions in a lab-controlled environment and should be able to cover most of the testing conditions. The proposed testing methodology consists of a deconstruction process which decomposes a domain into multiple test cases, and an aggregation

process which normalizes the measured KPIs and aggregates towards the domain level. An important pre-requisite before executing any test campaign is the calibration of the testbed, which includes the calibration of the measurements as well as the ETL process. Human panel experiments are introduced to fine tune the normalization functions and parameters to match with the human perception. A certain number of iterations is needed to obtain consistent results, while the exact number depends on the selected traffic profile and network scenario. A case study of QoE evaluation in the user experience domain with two use cases, namely high-speed internet and content streaming, is presented. The measurement results validated the effectiveness of the proposed testing methodology for QoE assessment of many applications and services. Furthermore, it is shown that the network scenarios have a great impact on the user experience, which provides some insight on how different channel conditions and radio resources will influence the overall QoE performance of an application.

## Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

## Data availability

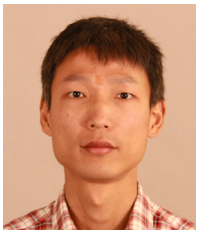
The data that has been used is confidential.

## Acknowledgments

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